

HARP: ExactFrac-Constrained Score Serving for GNN Prediction APIs

Krish Sharma

Academy of Math, Science, and Applied Technology
Ed W. Clark High School
Las Vegas, NV, USA
krish.sharma@unlv.edu

Shaikh Arifuzzaman

Department of Computer Science
University of Nevada, Las Vegas
Las Vegas, NV, USA
shaikh.arifuzzaman@unlv.edu

Abstract—GNN prediction APIs emit softmax scores consumed by ranking, routing, A/B tests, and compliance pipelines. Many such pipelines impose a concrete service-level requirement: a positive fraction of responses must be *replay-stable*—bitwise identical under re-query within a session—so caches hit, audit trails reproduce, and score-threshold decisions do not flicker. Any fresh uniform posterior perturbation (e.g., Laplace posterior perturbation, LBP) is infeasible under that requirement: every draw changes almost surely (ExactFrac=0). We propose HARP, a selective score-serving *framework* that keeps ExactFrac $\geq c$ by construction: a pluggable risk constructor ranks nodes, a hop expansion forms a protected set, and a pluggable protector (Laplace or masking) runs only there; Constrained Frac Search (CFS) then selects Frac to maximize Acc subject to ExactFrac $\geq c$ and an optional membership-audit cap LiRA $\leq \tau$. Release-only HARP is the default; topology ranking is a cheap constructor, not the core claim—audit-guided ranking is preferred when shadows are affordable. On citation and heterophilic graphs with five seeds, and on OGB serving-scale probes, HARP is feasible wherever $c > 0$ while uniform LBP is not; under $c = 0.60$ it preserves a clean majority with competitive accuracy and calibration, exposes Mass/Frac/session dials, and improves score-cache hit rates under fresh traffic. Membership audits guide deployment and expose residual clean-slice risk; HARP is not a claim of universal LiRA dominance or differential privacy.

Index Terms—graph neural networks, prediction APIs, ExactFrac SLA, selective release, score serving, data quality

I. INTRODUCTION

A. Motivation

GNN prediction APIs return softmax vectors for triage, ranking, and analytics [9], [10]. At scale the API is a continuous data product with Volume (graphs from 10^3 to 10^6 + nodes), Velocity (clients re-query), and Veracity (downstream systems need stable, well-calibrated scores). Released scores also leak training membership [1], [2], [4], [5].

a) Service-level constraint (ExactFrac).: We treat replay stability as an explicit SLA, not an aesthetic preference. A serving contract requires ExactFrac $\geq c$ (default $c = 0.60$ in our locked study): at least a c -fraction of node responses must be bitwise identical under re-query within a session. Operators

need this for (i) content-addressable score caches, (ii) deterministic audit replay of past decisions, and (iii) stable score-threshold pipelines (e.g., route if $p_{\text{top}} > \theta$). Fresh uniform LBP cannot meet any $c > 0$: each draw perturbs every coordinate almost surely (Remark 1). Session-sticky LBP ($B = 1$) restores cache hits but still yields ExactFrac = 0 across clients and fails audit-replay of *pre-session* answers.

b) Systems problem.:

$$\max_{\Pi} \text{Acc}(\Pi) \quad \text{s.t.} \quad \text{ExactFrac}(\Pi) \geq c, \quad \text{LiRA}(\Pi) \leq \tau$$

when the second constraint is active; otherwise maximize Acc subject to ExactFrac alone. Uniform mechanisms are infeasible for $c > 0$.

c) Framework.: **HARP** is a constrained score-serving framework with three interfaces: (1) a *constructor* that ranks nodes (topology, degree, train-neighbor exposure, confidence/entropy, random, or audit-guided vulnerability); (2) hop expansion $\mathcal{P}_k = \text{Ball}_k(\mathcal{R})$ plus a *protector* applied only on $\mathcal{P}_k$; (3) operator dials Frac/Mass/session budget B , with CFS searching Frac under (c, τ) . Default configuration is *release-only* selective Laplace at Frac = 0.40 (ExactFrac = 0.60).

d) Headline evidence.: Under ExactFrac ≥ 0.60 , HARP is feasible and uniform LBP is not; equal-mass LBP remains the right comparator when $c = 0$ is allowed and often wins Acc–LiRA in that unconstrained regime. Within the constrained class, constructor choice and CFS matter; residual clean-slice attacks are reported as first-class metrics. Serving benchmarks quantify cache hit rate, QPS, and latency under uniform/Zipf/bursty traffic.

B. Gap

Prior release defenses (LBP [5], MemGuard [6]) perturb every response. Train-time GNN MIA methods (GTD, MaskArmor [21], [22] and DP-GNNs [13], [14], [36]) do not expose ExactFrac/Mass serving accounting. None formalize ExactFrac $\geq c > 0$ as a hard API SLA with a constructor interface and constraint-aware search.

a) Running example (Cora, $c = 0.60$).: Equal-mass LBP ($b = 0.12$) at matched Mass reaches Acc ≈ 0.87 / LiRA ≈ 0.52 but ExactFrac = 0. Release-only HARP at Frac = 0.40 keeps ExactFrac = 0.60, Acc ≈ 0.83 , and much lower ECE on the

clean majority. Strong LBP ($b=0.3$) collapses Acc and remains ExactFrac=0—a secondary, not primary, baseline.

C. Contributions

- 1) **ExactFrac-constrained score serving.** Formalize Acc s.t. $\text{ExactFrac} \geq c$ (and optional $\text{LiRA} \leq \tau$); prove uniform LBP infeasible for $c>0$; motivate hop coverage in a one-hop linearized mixer.
- 2) **HARP framework + CFS.** Pluggable constructors and protectors; release-only default; CFS selects Frac under (c, τ) ; Mass/Frac/B telemetry.
- 3) **Constraint-aware evidence.** Feasibility tables and selective baselines (random/degree/confidence/audit/topology) at matched Frac; serving-bench cache/QPS/latency; clean-slice attack metrics; OGB serving probes with explicitly scoped audits.

Section II related work; Section III problem; Section IV HARP; Sections VI–VII evaluation; Section VIII discussion; Section IX conclusion.

II. BACKGROUND AND RELATED WORK

A. Membership inference on classical models

Shadow-model membership inference [1] and simple confidence features [2] show that prediction APIs leak training membership [3], [33]. Related extraction and inversion attacks [25], [26] further motivate treating scores as a privacy surface. Likelihood-ratio attacks (LiRA) are the current audit standard [4]. We use LiRA as an *operator audit* of whether release noise is warranted—not as the sole paper objective. We also track calibration because overconfident scores are a recurring membership cue [17], [34].

B. Release-time defenses on scores

MemGuard [6] adversarially perturbs confidence vectors to confuse attack classifiers while preserving the top-1 label; prediction purification similarly sanitizes released scores. Both act on *every* response and therefore cannot satisfy a clean-majority constraint. LBP [5] is the graph-native uniform Laplace baseline. HARP differs by selecting a hop-consistent minority and leaving the rest bit-exact.

C. Membership inference on graphs

Olatunji et al. [5] demonstrated node-level black-box MIAs on GNNs and proposed LBP and neighborhood sampling defenses. He et al. [38] systematically study node-level MIAs under varying topology knowledge. Subsequent work studies graph-level membership [7], embedding leakage [24], and broader attack taxonomies [8]. GTD [21] targets the transductive setting with alternate training and loss flattening. Khosla [39] shows that training-graph construction and inference-time edge access directly modulate membership advantage—orthogonal to our score-API audit protocol. GPS [35] studies structure $\rightarrow$ attribute leakage; we target score-based *membership*. Our threat model matches the score-based black-box node setting; we do not claim white-box, structural-modification, or federated attack coverage.

D. Architectures, regularization, and privacy

GCNs [9] apply spectral neighborhood averaging; GraphSAGE [10] uses mean aggregation and is often more robust under heterophily. DropEdge [11] randomly deletes edges during training. DP-SGD and DP-GNNs [13], [14] provide formal privacy guarantees at training time. Homophily modulates the train/test ϕ -gap [18]–[20]: under low h and sparsity, supervised memorization widens membership cues—motivating risk-weighted budget allocation.

E. Positioning

HARP is an ExactFrac-constrained *score-serving* framework for GNN APIs: utility and replay-stable fidelity are primary; membership audits are deployment dials. Uniform output noise (LBP, MemGuard) and train-time defenses (GTD, MaskArmor) are baselines or pluggable protectors. Formal DP (GAP) remains appropriate when a global ϵ guarantee is required [13]; we do not re-implement GAP under our splits. In short: LBP/MemGuard spend on every node (ExactFrac=0); GTD/MaskArmor harden training without ExactFrac accounting; GAP answers formal ϵ ; HARP answers constrained exact-fraction serving under optional audits.

a) *Evaluation scope.*: We separate serving-scale measurements (Acc, Mass/Frac, cache/QPS on large graphs) from computationally heavier privacy audits (LiRA shadow ensembles). Primary comparisons under $c>0$ are against other *selective* policies at matched Frac; equal-mass LBP is the honest unconstrained ($c=0$) reference.

III. PROBLEM SETUP

A. Notation

Let $G = (V, E)$ be an undirected graph with $n = |V|$ nodes, feature matrix $X \in \mathbb{R}^{n \times d}$, and labels $y \in \{0, \dots, C-1\}^n$. A random split partitions V into supervised train $\mathcal{T}$, validation $\mathcal{V}$, and test $\mathcal{Q}$ with ratios 40%/20%/40% (resampled per seed). Training is *transductive*: every node participates in message passing; only $\mathcal{T}$ enters the cross-entropy loss. We write $\hat{p}_v$ for the softmax posterior at v , r_v for a risk score in $[0, 1]$, $\mathcal{P}_k$ for the hop-expanded protected set, $\text{Mass} = \sum_v \sigma_v$, and $\text{Frac} = |\mathcal{P}_k|/n$.

B. Constrained score release

Operators rarely optimize Acc alone. They need usable accuracy *and* a reproducible majority of API responses, subject to a membership audit.

Definition 1 (Constrained Score Release). *Given a trained GNN and release channel, choose a release policy Π to*

$$\max_{\Pi} \text{Acc}(\Pi) \quad \text{s.t.} \quad \text{LiRA}(\Pi) \leq \tau, \quad \text{ExactFrac}(\Pi) \geq c,$$

where *ExactFrac* is the fraction of nodes whose released posterior equals $\hat{p}_v$ bit-exactly, $\tau \in [0.5, 1]$ is an audit threshold, and $c \in [0, 1]$ is the required clean fraction.

Remark 1 (Uniform LBP is infeasible for $c>0$). *Independent Laplace noise with scale $b>0$ on every coordinate changes the*

value almost surely, so $\text{ExactFrac} = 0$ for any uniform LBP policy. Selective release with protected fraction Frac achieves $\text{ExactFrac} = 1 - \text{Frac}$ by construction.

This reframes the Acc–LiRA comparison. Equal-mass LBP can win Acc subject to $\text{LiRA} \leq \tau$ when $c=0$ is allowed; HARP solves the strictly harder problem with $c>0$. The Frac sweep traces HARP’s Pareto frontier under that constraint.

C. Threat model

The adversary is a black-box client: it queries softmax scores and tries to guess whether a node was in the training set [2], [4], [5]. It does not see gradients or embeddings. White-box, link-stealing, and label-only APIs are out of scope [8], [12].

We also stress three practical variants: defense-aware shadows, multi-query averaging ($K \in \{1, 5, 20\}$), and an LTE-aware query adversary that focuses on clean nodes, protected-set boundaries, and top-decile LTE seeds (Sec. VII-C0k). We do not claim robustness to poisoning or unbounded queries without a session budget.

D. Attack features (brief)

Following Salem-style score attacks [2], we use confidence, true-class probability, entropy, and a margin term as ϕ -features. These are the signals membership audits and optional train-time alignment target.

E. Metrics

Utility: test accuracy (primary), plus F1/AUROC where useful. **Membership audit:** we report LiRA AUROC [4] and TPR at 0.1%/1% FPR—how well an attacker separates train members from held-out nodes (0.5 = chance). We also report confidence-based attacks for continuity with prior GNN work [2], [5]. **Data quality:** ECE [17], clean-slice ECE, and the clean-score majority ($1 - \text{Frac}$). LiRA tells operators *whether* to spend noise; Acc and ECE tell them *what they paid* in utility and score fidelity.

Remark 2. The primary protocol is transductive (non-members remain in the graph during training), matching common semi-supervised GNN APIs. We additionally report an inductive edge-exclusion check on Cora (Sec. VII-C0l) where test-incident edges are dropped at train/infer time.

IV. HARP: HOP-AWARE SELECTIVE RELEASE

A. Design principle

Uniform posterior noise cannot meet $\text{ExactFrac} \geq c > 0$ —any fresh $b > 0$ draw changes every response. Node-local selection without hop coverage is incomplete for GNNs: if v is protected but neighbor u is released exact, aggregation can reintroduce membership signal into u ’s score. HARP is a three-interface serving framework (Fig. 1):

- 1) **Constructor (pluggable):** rank nodes—topology (LTE), degree, train-neighbor exposure, confidence/entropy, random, or audit-guided vulnerability.

HARP: Hop-Aware Risk-conditioned Privacy



Fig. 1. HARP pipeline: risk ranking $\rightarrow$ high-risk seeds $\rightarrow$ k -hop expansion $\rightarrow$ protector only on the protected set.

- 2) **Geometry + protector:** expand $\mathcal{P}_k = \text{Ball}_k(\mathcal{R})$; apply Laplace or masking only on $\mathcal{P}_k$.
- 3) **CFS + dials:** search Frac under $\text{ExactFrac} \geq c$ and optional $\text{LiRA} \leq \tau$; log $\text{Mass}/\text{Frac}/B$.

Locked study config: *release-only* Laplace, $\text{Frac}=0.40$ ($\text{ExactFrac}=0.60$), $k=1$, $\sigma=0.30$. Topology LTE is the no-shadow default constructor; audit-guided ranking is preferred when shadows are already paid for. Train-time alignment is an optional ablation—not required for the ExactFrac claim.

a) *Noise mass and clean fraction.*: $\text{Mass} = \sum_v \sigma_v$ and $\text{Frac} = |\mathcal{P}_k|/n$ are operator accounting metrics. Uniform LBP at scale b has $\text{Mass}=bn$ and $\text{Frac}=1$. See paragraph IV-D0a for the locked-budget identity.

b) *Locked-config example (Cora).*: On Cora ($n=2708$), $\text{Frac}=0.40$ protects ≈ 1083 nodes after one-hop expansion; ≈ 1625 nodes stay exact. $\text{Mass}_{\text{HARP}} \approx 325$ matches equal-mass LBP with $b=0.12$. Strong LBP ($b=0.3$) spends $\text{Mass} \approx 812$ on all nodes—the Acc-collapse baseline in Table II.

c) *Release transform (Laplace protector).*: Let $\sigma_v = \sigma_{\text{strong}} \mathbb{1}[v \in \mathcal{P}_k]$. The released posterior is

$$\tilde{p}_v = \text{renorm}([\hat{p}_v + \text{Lap}(0, \sigma_v)]_+), \quad (1)$$

with independent Laplace coordinates and the same transform on every defense-aware shadow. The masking protector instead returns $\hat{p}_v$ for $v \notin \mathcal{P}_k$ and a one-hot of $\arg \max \hat{p}_v$ for $v \in \mathcal{P}_k$.

d) *Algorithm (systems view).*:

- 1) **Risk:** $r \leftarrow \text{Constructor}(G, \mathcal{T})$ (topology / degree / confidence / random / audit).
- 2) **CFS (optional):** search $\text{Frac} \leq 1 - c$ to maximize Acc subject to $\text{LiRA} \leq \tau$.
- 3) **Seeds + expand:** binary-search seeds so $|\text{Ball}_k(\mathcal{R})|/n \approx \text{Frac}$; $\mathcal{P}_k \leftarrow \text{Ball}_k(\mathcal{R})$.
- 4) **Release:** apply (1) (or masking); log ($\text{Mass}, \text{Frac}, \text{Acc}, \text{LiRA}, \text{ECE}$).

Train-time alignment is optional and omitted in the locked release-only configuration.

B. Risk constructors

The constructor interface is part of the framework contribution. **Topology (LTE)** uses only G , $y|\mathcal{T}$, and the train mask (inverse degree, local heterophily, train-neighbor exposure); it is a cheap default when shadows are unavailable—not claimed to be privacy-optimal. **Degree / train-neighbor** isolate individual topology cues. **Confidence / entropy** rank from a clean

forward pass (high max-confidence or high entropy). **Random** is the matched-Frac selectivity control. **Audit-guided** scores $|\mu_{\text{IN}} - \mu_{\text{OUT}}|$ over a small shadow ensemble [37] and is preferred when that ensemble is already paid for. We report Spearman correlation between each constructor and audit vulnerability (Table XIII) and Acc/LiRA at matched Frac (Table XI). On Cora/Citeseer, *entropy* is the strongest cheap prior (Spearman ≈ 0.33 – 0.35); topology (LTE) is moderate (≈ 0.25 – 0.28); max-confidence is anti-correlated. Heterophilic Chameleon priors are weak—selectivity still enforces Exact-Frac, but ranking gains are limited. Where topology underperforms random on LiRA (e.g., Cora in Table XI), we state that plainly: ExactFrac comes from selectivity, not from LTE.

C. Why k -hop expansion

Let $\mathcal{R}$ be seeds. A GNN score at u already mixes information from $\text{Ball}_k(u)$ before any release noise is applied. If a membership-sensitive seed $v \in \mathcal{R}$ is noised at release but a neighbor u is returned bit-exact, the adversary simply queries u .

Proposition 1 (Hop coverage under a one-hop linear mixer). *Fix a one-hop score model $s_u = \alpha z_v + (1 - \alpha) z_u$ for $u \in \mathcal{N}(v)$, $\alpha \in (0, 1]$, where z_v carries a membership cue of gap $\gamma > 0$, and release adds independent Laplace noise only to chosen nodes’ scores. If the policy noises v but leaves u exact, an adversary observing s_u retains the unprotected leaf advantage. Noising $\text{Ball}_1(\{v\})$ places Laplace noise on every score that linearly depends on z_v under this mixer.*

Proof sketch. Under seed-only protection the released leaf equals the clean mixer output, so the member/non-member gap in $\mathbb{E}[s_u]$ remains $\alpha\gamma$. Expanding protection to u subjects that coordinate to $\text{Lap}(0, \sigma)$. A synthetic mixer confirms the ordering: unprotected and seed-only leaf AUROCs stay near 1.0, while hop-1 protection lowers the leaf attack to ≈ 0.72 (Fig. 6, left). $\square$

This is a motivating linearized result—not a necessity theorem for nonlinear multi-layer GNNs. Empirically we set k to match the model’s effective receptive field (default $k=1$ for 2-layer SAGE/GCN; $k \in \{0, 2\}$ in Table XII). Frac—not k —is the population Acc–Mass knob.

D. Complexity and accounting

Proposition 2 (HARP setup and release complexity). *Let $G = (V, E)$ have $n = |V|$ nodes and $m = |E|$ adjacency endpoints, with C classes. Fix hop radius $k = O(1)$ and $T = O(1)$ binary-search iterations for a target protected fraction. Then setup (LTE + seeds + k -hop expand) runs in $O(m + n \log n)$ time and $O(n + m)$ space. Release runs in $O(nC)$ —matching uniform LBP’s tier, with less noise-sampling work whenever $\text{Frac} < 1$.*

Proof sketch. LTE scans edges then normalizes n scores: $O(m+n)$. Seed selection is $O(n \log n)$ plus T BFS expansions at $O(m+n)$. Release writes scales, samples Laplace on $\mathcal{P}_k$, and renormalizes: $O(nC)$. $\square$

Remark 3 (Per-response Laplace; no global DP claim). *Interpret each node’s clean posterior as a query with ℓ_1 -sensitivity at most $\Delta=2$ on the simplex. Independent $\text{Lap}(\sigma)$ on coordinates of $v \in \mathcal{P}_k$ is a per-response Laplace mechanism with scale σ ; unprotected nodes receive no noise. Any policy with $\text{ExactFrac} > 0$ has infinite global (ε, δ) -DP by construction. We report this only as an accounting remark—not as a local-DP guarantee under graph adjacency—and recommend training-time DP (e.g., GAP) when a legal global ε is required [13].*

a) *Accounting identity (stated once).*: Under locked Frac and fixed σ_{strong} , $\text{Mass}_{\text{HARP}} = \text{Frac} \cdot n \cdot \sigma_{\text{strong}}$ by construction. Mass is an operational dial correlated with mean ℓ_1 distortion and top-1 flip rate (Table XV); it is not a standard privacy budget. Empirical claims are (i) feasibility under $\text{ExactFrac} \geq c$, (ii) Acc/ECE within the selective class at matched Frac/Mass, and (iii) serving telemetry.

E. What HARP is not

HARP is not differential privacy and is not a claim of lower population LiRA than uniform noise under $\text{ExactFrac} = 0$. It is not a new Laplace mechanism. It is a constrained ExactFrac serving framework: spend a protector on a hop-expanded subset, keep the rest replay-stable, search Frac with CFS, and audit residual risk—including on unprotected nodes.

a) *Intuition.*: Think of HARP as a two-tier API under an ExactFrac SLA. Unprotected nodes return the model’s exact softmax (replay-stable). Protected nodes (and their k -hop neighborhood) receive the chosen protector. The constructor ranks where to spend; Frac/CFS set how much; B limits query averaging for authenticated sessions.

V. MOTIVATION: WHEN SELECTIVE PRIVACY PAYS

Matched-budget undefended LiRA (Table XVIII) concentrates in small/mid heterophilic cells; several mid/large graphs sit near chance under $n_{\text{shadows}}=4$. Audit first: do not pay strong uniform noise where $\text{LiRA} \approx 0.5$; deploy HARP where $\text{LiRA} \gg 0.5$ or ExactFrac is required under a strong- σ policy. Heterophilic graphs (Chameleon/Actor) show auditable LiRA; Cora/Citeseer emphasize veracity at matched Mass; PubMed/Photo/ogbn often warrant Frac=0 unless policy forces strong σ .

VI. EXPERIMENTAL PROTOCOL

A. Pre-declared rules

Primary objective: Acc under $\text{ExactFrac} \geq c$ (locked $c=0.60$). **Primary metrics:** Acc, ExactFrac, ECE (fidelity). **Audit metric:** LiRA AUROC (and clean-slice confidence AUROC) as deployment constraints—not the headline win condition. **Operational metrics:** Mass, Frac, cache hit rate / QPS. Hyperparameters are not selected to minimize test LiRA. Under $c>0$, primary baselines are selective policies at matched Frac; equal-mass LBP is the unconstrained ($c=0$) reference; strong LBP ($b=0.3$) is a secondary published recipe showing Acc collapse under full-graph strong noise.

B. Systems metrics (definitions)

- **ExactFrac**: fraction of nodes whose released posterior equals $\hat{p}_v$ bit-exactly.
- **Mass** = $\sum_v \sigma_v$: operational Laplace-scale spend (correlated with distortion; Table XV).
- **Frac** = $|\mathcal{P}_k|/n$: protected fraction (ExactFrac=1 – Frac under weak=0).
- **ECE**: calibration of released probabilities.

C. Locked HARP (release-only)

Release-only selective Laplace: $\lambda=0$, no HCAG, no train-on-protected; $k=1$, $\sigma_{\text{strong}}=0.30$, Frac=0.40 (ExactFrac=0.60). Constructor default in locked tables: topology (LTE); audit/random/degree/confidence appear in Table XI. Train-time alignment is reported only as an ablation. Baselines: equal-mass LBP $b=0.12$; strong LBP $b=0.3$; MemGuard; GTD; MaskArmor; selective masking; DP-SGD Acc-tuned (vacuous ε).

D. Baselines and attacks

Gaussian LiRA with $n_{\text{shadows}}=4$ (primary); sweeps $\{4, 16, 64\}$ on Cora/Chameleon. Five seeds with mean $\pm 95\%$ CI on primary Cora/Chameleon/Citeseer constructor tables; paired seeds across defenses. ogbn-arxiv: NeighborLoader serving Acc with LiRA at $n_{\text{shadows}} \in \{2, 4\}$ (audit stress, not a million-node privacy claim). ogbn-products: full-graph serving RSS/QPS; LiRA only on a labeled BFS 15k-node probe.

E. What each baseline tests

Equal-mass LBP ($b=0.12$) matches HARP Mass at Frac=0.40 and is the unconstrained ($c=0$) reference. **Selective constructors** (random/degree/confidence/entropy/audit/topology) isolate ranking quality at matched Frac under the SLA. **Strong LBP** ($b=0.3$) is a secondary published recipe that collapses Acc under full-graph strong noise. **MemGuard** is an iterative L1-ball release defense with an MLP attack model (hardened vs. our earlier random-search stub; still ExactFrac=0). **GAP-agg** is a GAP-inspired aggregation-Laplace baseline with analytical (ε, δ) accounting under our splits (not a full GAP codebase reproduction). **Selective masking** is an alternate protector (ExactFrac preserved; often worse LiRA). **GTD/MaskArmor** are train-time defenses (no release Mass/Frac). **Clip+noise DP-SGD** remains an Acc-tuned vacuous- ε reference.

F. Attack protocol (summary)

We train defense-aware shadow models, release transformed posteriors, and report LiRA AUROC with ϕ -features [2], [4]. Validation nodes are excluded from membership labels; test Acc/ECE use the held-out test split. Multi-query experiments average $K \in \{1, 5, 20\}$ independent draws; session policies cap fresh draws at $B=5$.

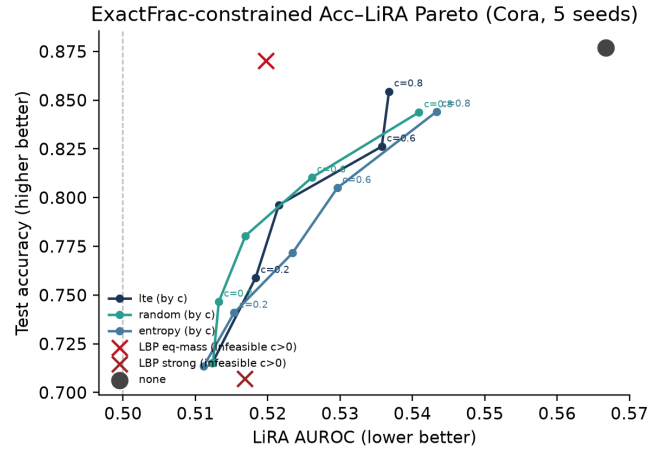


Fig. 2. ExactFrac-constrained Acc-LiRA Pareto on Cora (5 seeds). Curves: selective HARP constructors at $c \in \{0.2, 0.4, 0.6, 0.8\}$ (ExactFrac= c). Uniform LBP markers are ExactFrac=0 and infeasible for every $c>0$.

G. Negative controls

PubMed (near-chance undefended LiRA): selective privacy should not be forced to spend a full uniform budget; HARP still preserves Acc vs. strong LBP at Frac=0.40 (Acc 0.839 vs. 0.768). ogbn-arxiv / ogbn-products: near-chance or memory-bounded audits under matched budgets—audit first, harden second [28].

VII. RESULTS

A. Overview

a) *Operational headline.*: Under ExactFrac ≥ 0.60 : (i) HARP is feasible and uniform LBP is not (Table I); (ii) release-only HARP keeps Acc competitive within the selective class while returning a 60% replay-stable majority; (iii) serving benchmarks show content-cache hits from ExactFrac under fresh traffic (Table XIV); (iv) equal-mass LBP can still win Acc-LiRA when $c=0$ is allowed—that unconstrained regime is out of scope for the SLA.

Questions: (Q1) Feasibility under $c>0$? (Q2) Acc/ECE vs. selective and equal-mass baselines? (Q3) Do constructors beat random? (Q4) Serving cost and clean-slice residual risk? (Q5) OGB serving scale with scoped audits?

B. How to read the trade space

Pick the constraint first. If ExactFrac $\geq c>0$ is required, uniform rows are marked infeasible. If only Acc subject to LiRA $\leq \tau$ matters, prefer equal-mass LBP or MemGuard. HARP targets the first setting.

C. HARP under ExactFrac ≥ 0.60 : accuracy and score quality

a) *Primary claim.*: Fig. 3 and Table II: under ExactFrac ≥ 0.60 , release-only HARP is feasible; equal-mass LBP and MemGuard are not. At matched Mass ≈ 325 , HARP yields ECE 0.033 with a 60% replay-stable majority; equal-mass ECE 0.159 with ExactFrac=0. Equal-mass can win Acc-LiRA when the SLA is dropped—that is expected and out of the constrained objective.

TABLE I
FEASIBILITY UNDER EXACTFRAC $\geq c$ (DESIGN EXACTFRAC). UNIFORM LBP IS INFEASIBLE FOR EVERY $c > 0$; SELECTIVE HARP MEETS c BY SETTING FRAC = $1 - c$.

Policy	$c=0$	$c=0.2$	$c=0.6$	$c=0.8$
none	yes	yes	yes	yes
LBP (any $b > 0$)	yes	no	no	no
LBP + session $B=1$	yes*	no†	no†	no†
HARP (Frac = $1 - c$)	yes	yes	yes	yes

*Sticky sessions restore cache hits but ExactFrac across clients remains 0.

†Bit-exactness fails for fresh clients and pre-session audit replay.

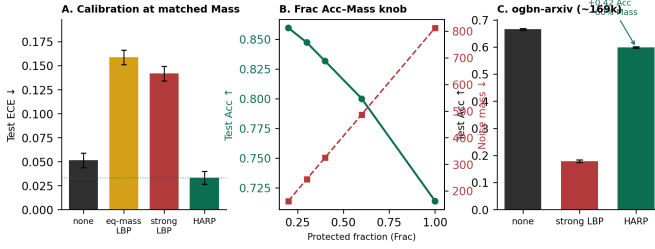


Fig. 3. Systems headline. (A) ECE at matched Mass. (B) Frac Acc-Mass dial. (C) ogbn-arxiv Acc vs. strong LBP.

TABLE II
CORA GRAPH SAGE UNDER EXACTFRAC SLA $c=0.60$ (5 SEEDS). FEASIBLE ROWS ONLY FOR $c > 0$: HARP / HARP-MASK. EQUAL-MASS WINS ACC-LIRA WHEN $c=0$ IS ALLOWED (EXACTFRAC=0). †GAP-AGG ANALYTICAL ε (GAP-INSPIRED; NOT A FULL GAP REPRODUCTION). ‡TRAIN-TIME ONLY (NO RELEASE MASS/FRAC). §HARDENED ITERATIVE MEMGUARD (MLP ATTACK; EXACTFRAC=0).

Defense	Feas.	Acc	LiRA	ECE↓
none	yes	0.877	0.567	0.051
LBP equal-mass	no	0.870	0.520	0.159
MemGuard§	no	0.877	0.617	0.067
LBP $b=0.3$	no	0.707	0.517	0.142
GAP-agg ($\sigma=3$, $\varepsilon \approx 41$)†	no	0.776	0.519	0.399
GAP-agg ($\sigma=5$, $\varepsilon \approx 19$)†	no	0.442	0.500	0.287
GTD‡ / MaskArmor‡	—	0.878 / 0.877	0.538 / 0.580	0.091 / 0.123
HARP (release)	yes	0.826	0.536	0.033
HARP-mask	yes	0.885	0.639	0.092

b) Secondary: Acc vs. published strong LBP. Against LBP $b=0.3$ [5], HARP raises Acc $0.707 \rightarrow 0.832$ at Frac=0.40 because strong noise applies to a minority of nodes (Tables II, IV). This is mechanically expected and is not the constrained-feasibility claim.

c) Membership audit (deployment dial). At matched Mass, equal-mass LBP can beat HARP on Acc-LIRA (Cora Acc 0.870 / LiRA 0.520). If ExactFrac is not required, prefer equal-mass LBP. Under the SLA, LiRA guides whether Frac must rise and whether the audit threshold τ is met after CFS.

GTD and MaskArmor rows report Acc/LIRA/ECE only: both defenses modify training, not release posteriors, so they have no Mass or Frac to compare against HARP’s release budget. MemGuard preserves Acc but leaves ExactFrac=0 and, under our hardened iterative attack model, does not reduce LiRA relative to undefended. GAP-agg provides a finite analytical ε under our splits but forces ExactFrac=0

TABLE III
CLEAN-SLICE ECE UNDER LOCKED HARP (MULTI-DATASET MEANS; CORA ALSO MEASURED AT 5 SEEDS).

Dataset	ECE_{clean}	ECE_{prot}	ECE_{pop}	none ECE
Cora	0.020	0.090	0.037	0.053
Citeseer	0.083	0.121	0.094	0.140
Chameleon	0.273	0.186	0.238	0.290
Actor	0.254	0.277	0.262	0.262

TABLE IV
 ΔAcc (HARP—STRONG LBP) UNDER LOCKED FRAC=0.40.

Dataset	ΔAcc	Mass _{LBP}	Mass _{HARP}	Frac
Cora	+0.125	812	325	0.40
Citeseer	+0.096	998	399	0.40
Chameleon	+0.061	683	273	0.40
Actor	+0.040	2280	912	0.40
PubMed	+0.070	5915	2366	0.40
Photo	+0.185	2295	946	0.40

TABLE V
MATCHED-MASS: EQUAL-MASS LBP ($b=0.12$) VS. RELEASE-ONLY HARP (CORA, 5 SEEDS).

Dataset	HARP Acc	Eq Acc	HARP ECE	Eq ECE	Clean?
Cora	0.826	0.870	0.033	0.159	HARP 60%
Citeseer	0.716	0.740	0.094	0.073	HARP 60%
Chameleon	0.452	0.439	0.238	0.214	HARP 60%
Actor	0.327	0.324	0.262	0.244	HARP 60%

and trades Acc for ε (Table II). Selective masking keeps ExactFrac=0.60 and Acc high, but raises LiRA—justifying the Laplace protector when the audit requires it.

—Slice ECE localizes the win on Cora (5 seeds): $ECE_{\text{clean}}=0.018$ vs. $ECE_{\text{prot}}=0.081$ (population 0.033). Clean nodes match undefended calibration; protected nodes carry the noise penalty. HARP’s population ECE can beat undefended none because train-time ϕ -alignment/HACG reduces overconfidence on the clean majority; on Chameleon, HARP Acc can slightly exceed undefended Acc for the

same reason. Across datasets, the portable quality property is the clean majority and clean-slice ECE (Table III). Equal-mass population ECE can look competitive when weak noise regularizes overconfidence (Table V), but every response is still perturbed.

d) Multi-dataset grid. Table VI repeats the Acc/LIRA pattern on six graphs; PubMed/Photo are near-chance LiRA controls where HARP still preserves Acc vs. strong LBP. Reading down each triple: HARP consistently recovers 4–19 Acc points vs. strong LBP while LiRA moves modestly (often +0.01–+0.02 vs. strong LBP, still below undefended). At the security-relevant low-FPR point, HARP also cuts TPR@1% FPR relative to undefended (Cora 0.033 $\rightarrow$ 0.011; Chameleon 0.059 $\rightarrow$ 0.011; Citeseer 0.042 $\rightarrow$ 0.014). The heterophilic rows (Chameleon, Actor) show the largest undefended LiRA and the clearest case for spending a Frac budget; the near-chance rows (PubMed, Photo) show that HARP’s Acc benefit does not require a population LiRA win—operators audit first,

TABLE VI
HARP PRIMARY GRID (GRAPHSAGE; CORA 5-SEED, OTHERS 3-SEED).
TPR = LiRA TPR@1% FPR.

Dataset	Defense	Acc	LiRA	TPR	Mass	Frac
Cora	none	0.877	0.567	0.033	—	—
	LBP $b=0.3$	0.707	0.517	0.019	812	1.00
	HARP	0.832	0.533	0.011	325	0.40
Citeseer	none	0.744	0.591	0.042	—	—
	LBP $b=0.3$	0.620	0.523	0.030	998	1.00
	HARP	0.716	0.548	0.014	399	0.40
Chameleon	none	0.448	0.618	0.059	—	—
	LBP $b=0.3$	0.391	0.539	0.038	683	1.00
	HARP	0.452	0.555	0.011	273	0.40
Actor	none	0.335	0.610	0.037	—	—
	LBP $b=0.3$	0.287	0.549	0.032	2280	1.00
	HARP	0.327	0.561	0.019	912	0.40
PubMed	none	0.876	0.505	0.012	—	—
	LBP $b=0.3$	0.768	0.499	0.010	5915	1.00
	HARP	0.839	0.502	0.012	2366	0.40
Photo	none	0.802	0.500	—	—	—
	LBP $b=0.3$	0.552	0.499	—	2295	1.00
	HARP	0.737	0.501	—	946	0.40

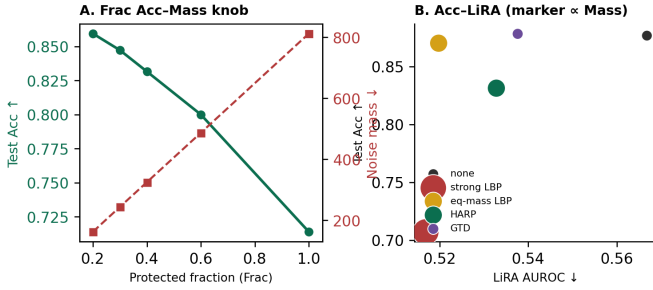


Fig. 4. Frac dial (left) and Acc–LiRA tradeoff with marker size $\propto$ Mass (right). Equal-mass LBP and GTD shown for context.

TABLE VII
HARP FRAC SWEEP ON CORA (5 SEEDS; $\sigma_{\text{strong}}=0.30$).

Frac	Acc $\uparrow$	LiRA $\downarrow$	Mass $\downarrow$
0.20	0.859 [0.855, 0.866]	0.537 [0.532, 0.543]	162
0.30	0.847 [0.843, 0.853]	0.538 [0.533, 0.543]	244
0.40	0.832 [0.826, 0.839]	0.533 [0.525, 0.541]	325
0.60	0.800 [0.793, 0.809]	0.519 [0.512, 0.526]	487
1.00	0.714 [0.699, 0.727]	0.511 [0.503, 0.519]	812

then decide whether $\text{Frac} > 0$ is warranted.

e) *Frac dial and Mass accounting.*: Fig. 4 and Table VII sweep Frac on Cora (5 seeds): Acc falls and LiRA improves as Frac rises. Frac is the Acc–Mass dial under fixed σ_{strong} —the systems analog of an empirical privacy spend knob.

f) *Constrained Frac Search.*: Given SLA $c=0.60$ and audit cap $\tau=0.55$, CFS searches $\text{Frac} \in \{0, 0.2, 0.3, 0.4\}$ and returns the Acc-maximizing feasible point (Table VIII). On Cora (3 seeds), CFS typically selects $\text{Frac}=0.20$ ($\text{ExactFrac}=0.80 > c$) with $\text{Acc} \approx 0.85$ —strictly better Acc than the locked $\text{Frac}=0.40$ point while still meeting τ . Locked $\text{Frac}=0.40$ remains the conservative operating point when operators want a fixed dial; CFS is the constraint-aware allocator.

g) *Exact scores $\neq$ membership safety.*: Clean nodes are well calibrated ($\text{ECE}_{\text{clean}}=0.018$), but they are not auto-

TABLE VIII
CFS ON CORA UNDER $\text{EXACTFRAC} \geq 0.60$ AND $\text{LiRA} \leq 0.55$ (3 SEEDS;
MEAN SELECTED $\text{FRAC}/\text{ACC}/\text{LiRA}$).

Constructor	Selected Frac	Acc	LiRA
topology (LTE)	0.20	0.856	0.539
random	0.23	0.841	0.536
audit	0.23	0.848	0.540
entropy	0.27	0.834	0.542

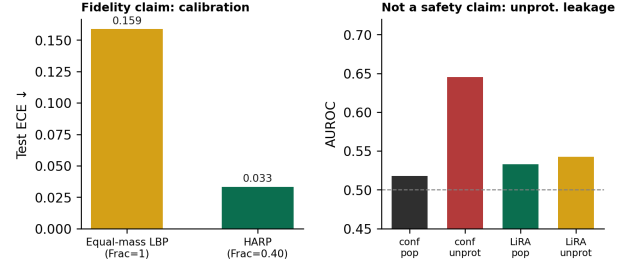


Fig. 5. Veracity vs. safety on Cora. Left: ECE at matched Mass. Right: unprotected-slice confidence AUROC remains 0.645.

TABLE IX
MULTI-QUERY LiRA ON CORA GRAPHSAGE (5 SEEDS; $n_{\text{shadows}}=4$).

Defense	K=1 LiRA	K=5 LiRA	K=20 LiRA
none	0.567	0.567	0.567
HARP	0.536	0.546	0.551
eq-mass LBP	0.520	0.520	0.520

TABLE X
SESSION BUDGET ON HARP CORA AT $K=20$ (5 SEEDS). $B=1$ FREEZES
ACC/ECE NEAR THE ONE-SHOT RELEASE; $B=20$ MATCHES UNCAPPED
AVERAGING.

Policy	B	Acc	LiRA
Budget reuse	1	0.832	0.540
Budget reuse	5	0.851	0.540
Budget reuse	20	0.885	0.545
Uncapped avg	—	0.886	0.545

matically hard to attack. Under locked HARP, unprotected-slice confidence AUROC is ≈ 0.64 vs. ≈ 0.51 population-wide (Fig. 5; Table XVII). Raise Frac or enforce session budget B if clean-slice canaries drift up. Aerni et al. [37] motivate such canary checks (Table XX(a)).

h) *Velocity: multi-query and session budgets.*: Averaging K independent releases denoises Laplace and moves toward the undefended baseline (Table IX). A session policy of at most B fresh draws per *authenticated principal*—reusing the last draw thereafter—caps this drift (Table X). **Enforcement assumption:** server-side state bound to authenticated identities with rate limits. Sybil-capable attackers who mint many principals can bypass B ; we do not claim Sybil resistance. $B=1$ is the default when velocity is adversarial under authenticated sessions; Frac and B are complementary controls.

i) *Component ablations (consolidated).*: Table XI answers constructor quality under the ExactFrac SLA (5-seed Cora). Table XII keeps hop/release-only ablations. Under

TABLE XI

RELEASE-ONLY SELECTIVE CONSTRUCTORS AT $\text{FRAC}=0.40$ / $\text{EXACTFRAC}=0.60$ (CORa, 5 SEEDS; $\text{MEAN} \pm 1.96 \text{ SE}$). EQUAL-MASS WINS ACC—LiRA ONLY WHEN $\text{EXACTFRAC}=0$ IS ALLOWED.

Policy	Feas. $c=0.6$	Acc	LiRA	ExactFrac
none	yes	0.877 ± 0.005	0.567 ± 0.006	1.00
LBP equal-mass	no	0.870 ± 0.005	0.520 ± 0.011	0.00
LBP $b=0.3$	no	0.707 ± 0.013	0.517 ± 0.011	0.00
sel. audit	yes	0.820 ± 0.010	0.526 ± 0.009	0.60
sel. random	yes	0.810 ± 0.006	0.526 ± 0.007	0.60
sel. entropy	yes	0.805 ± 0.010	0.530 ± 0.014	0.60
sel. topology (LTE)	yes	0.826 ± 0.006	0.536 ± 0.008	0.60
sel. train-nbr	yes	0.838 ± 0.006	0.537 ± 0.008	0.60
sel. degree	yes	0.819 ± 0.010	0.543 ± 0.007	0.60
sel. max-conf	yes	0.813 ± 0.016	0.550 ± 0.012	0.60

TABLE XII

HOP/RELEASE ABLATIONS AT MATCHED $\text{FRAC} \approx 0.40$ (CORa, 5 SEEDS, RELEASE-ONLY).

Cora variant	Acc	LiRA	Mass	ExactFrac
LBP $b=0.3$	0.707	0.517	812	0.00
HARP release $k=0$	0.829	0.524	325	0.60
HARP release $k=1$	0.826	0.536	325	0.60
HARP release $k=2$	0.817	0.530	325	0.60

TABLE XIII

SPEARMAN(r , AUDIT VULNERABILITY), 5 SEEDS. ENTROPY IS THE STRONGEST CHEAP PRIOR ON CORa/CITeseer; TOPOLOGY IS MODERATE; MAX-CONF ANTI-CORRELATED.

Dataset	Constructor	Spearman	95% CI
Cora	entropy / LTE / degree	0.33 / 0.25 / 0.13	see CSV
Citeseer	entropy / LTE	0.35 / 0.28	see CSV
Chameleon	max-conf / LTE	0.12 / -0.07	weak

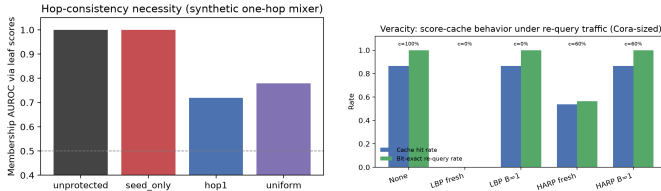


Fig. 6. Left: hop necessity (Prop. 1)—seed-only leaves leaf AUROC at 1.0; hop-1 lowers it to ≈ 0.72 . Right: score-cache hits under re-query traffic (Table XVI).

the SLA, $\text{audit} \approx \text{random}$ on LiRA on Cora; topology (LTE) is not privacy-optimal there. On heterophilic Chameleon (5 seeds), train-neighbor and topology beat random on LiRA ($0.559 / 0.563$ vs. 0.572) while keeping $\text{ExactFrac}=0.60$ —ranking helps when structure correlates with leakage. Release-only recovers most Acc—selectivity, not train-time alignment, carries ExactFrac .

j) *Why ExactFrac matters (serving benchmark).*: Beyond the synthetic ExactFrac corollary (Table XVI), we run an in-process Cora score API under uniform, Zipf, and bursty workloads (Table XIV): content-addressable caches key on bit-exact posteriors, so fresh LBP never hits; HARP fresh hits on the unprotected majority; session $B=1$ restores hits for sticky clients but does not create ExactFrac for new clients.

TABLE XIV

IN-PROCESS SERVING BENCH (CORa; 20K QUERIES; APPLE M4 CPU). CONTENT-ADDRESSABLE KEYS ON BIT-EXACT POSTERIORIS.

Workload	Policy	Hit	QPS	p50 ms	p95 ms
zipf	none	0.937	4.0×10^6	<0.001	<0.001
zipf	lbp_fresh	0.000	9.3×10^3	0.104	0.116
zipf	harp_fresh	0.606	6.2×10^5	<0.001	0.004
zipf	lbp_B1	0.937	1.5×10^5	<0.001	0.101
zipf	harp_B1	0.937	2.2×10^6	<0.001	<0.001
uniform	harp_fresh	0.515	5.3×10^5	<0.001	0.004
bursty	harp_fresh	0.594	5.6×10^5	<0.001	0.004

TABLE XV

MASS VS. DISTORTION ON CORa (5 SEEDS). MATCHED $\text{MASS} \approx 325$; HARP LOWERS $\text{mean } \ell_1/\text{JS}$ ON AVERAGE BY CONCENTRATING NOISE; EXACTFRAC IS BY DESIGN.

Policy	Mass	mean ℓ_1	top-1 flip	ExactFrac
LBP equal-mass	325	0.493	0.017	0.00
HARP release	325	0.349	0.088	0.60

TABLE XVI

SCORE-CACHE SIMULATION (CORa-SIZED; $\text{EXACTFRAC } c$ BY DESIGN) ALONGSIDE THE SERVING BENCH.

Policy	c	Hit rate	Bit-exact re-query
None	1.00	0.865	1.000
LBP fresh	0.00	0.000	0.000
LBP $B=1$	0.00	0.865	1.000
HARP fresh	0.60	0.537	0.563
HARP $B=1$	0.60	0.865	1.000

TABLE XVII

LTE-AWARE ADAPTIVE QUERIES ON CORa HARP (5 SEEDS).

Query set	Conf@0.4	Conf@0.6	LiRA@0.4
Population	0.518	0.473	0.533
Unprotected (clean)	0.645	0.633	0.542
Boundary (clean)	0.634	0.628	0.538
Top-decile LTE	0.641	0.511	0.529

ExactFrac is also required for audit replay and deterministic score-threshold routing—not only caching.

k) *Clean-slice residual risk (first-class).*: Unprotected nodes are *not* claimed safe. Table XVII: querying unprotected, boundary, or top-decile topology seeds raises conf AUROC $0.518 \rightarrow 0.634\text{--}0.645$ under locked HARP, while population LiRA stays ≈ 0.53 . HARP offers configurable risk reduction under an ExactFrac SLA; raise Frac , switch to audit-guided ranking, and/or set $B=1$ until both population LiRA and worst-slice conf AUROC meet operator thresholds.

l) *Inductive edge exclusion.*: Dropping test-incident edges (Cora; 3 seeds) raises undefended LiRA to 0.631 (TPR@1% FPR 0.036). HARP recovers Acc $0.486 \rightarrow 0.620$ vs. strong LBP at $\text{Frac}=0.40$ (LiRA 0.569; ECE 0.037). At $\text{Frac}=0.60$, Acc falls to 0.549 while LiRA improves to 0.559—inductive audits may warrant a higher Frac on the same dial.

TABLE XVIII
AUDIT CONTROLS (GRAPHSAGE; 3 SEEDS). *HETEROPHILIC.
(B) COMPARATIVE SHADOW SCALING (NOT HARP-ONLY).

(a) Undefended LiRA ($n_{sh}=4$)			(b) Cora shadow scaling (LiRA / TPR@1%)			
Dataset	n	Acc	LiRA	none	LBP	HARP
Chameleon*	2 277	0.449	0.623	0.568 / 0.033	0.517 / 0.022	0.534 / 0.057
Cora	2 708	0.879	0.668	0.808 / 0.201	0.555 / 0.008	0.670 / 0.057
Actor*	7 600	0.335	0.610	0.847 / 0.265	0.561 / 0.008	0.677 / 0.073
Photo	7 650	0.825	0.500	Acc-invariant: none 0.879, LBP 0.716, HARP 0.505		
PubMed	19 717	0.876	0.505	0.836. Chameleon mirrors the pattern (at $n_{sh}=64$: none 0.965, LBP 0.685, HARP 0.811; Δ Acc vs. LBP +0.061).		
ogbn-arxiv	169 343	0.666	0.503			

TABLE XIX
OGBN-PRODUCTS SERVING PROBE + BFS 15K LiRA ($n_{sh}=2/4$).

Setting	n	Acc	LiRA
Load / train none	2.45M	0.695	—
LBP / HARP Mass	2.45M	—	—
none / LBP / HARP ($n_{sh}=2$)	15k	0.874/0.343/0.686	0.506/0.500/0.503
none / LBP / HARP ($n_{sh}=4$)	15k	0.875/0.346/0.693	0.515/0.503/0.510

D. When to spend: volume $\times$ structure

Table XVIII(a): undefended LiRA concentrates in small/mid heterophilic cells; Photo/PubMed/ogbn-arxiv sit near chance under matched shadows—audit first, then set Frac. Table XVIII(b) reports the honest shadow-count story on Cora for none / LBP / HARP. Absolute LiRA rises with $n_{shadows}$ for every defense [4]; Acc is invariant. HARP does *not* beat strong LBP on LiRA at large shadow counts (Cora $n_{sh}=64$: LBP 0.561 vs. HARP 0.677)—expected when ExactFrac=0 is allowed. What HARP keeps is (i) strictly lower LiRA than undefended and (ii) Acc recovery vs. strong LBP (Δ Acc +0.12, invariant in n_{sh}). TPR@1% FPR follows the same pattern.

E. Canaries and large-graph volume

Table XX(a): HARP suppresses planted/top-decile confidence cues relative to none/LBP; population LiRA remains primary. Table XX(b): on ogbn-arxiv ($\sim 169k$; NeighborLoader; gate off, warmup 3) [28], strong LBP drops Acc to 0.179 [0.175, 0.184]. HARP recovers 0.599 [0.597, 0.601] (paired Δ Acc +0.420 [0.414, 0.425]; 3 seeds) at Frac=0.40. Undefended LiRA stays near chance at $n_{sh} \in \{2, 4\}$ and also at $n_{sh}=16$ (LiRA 0.510; Acc 0.666)—so the Acc recovery is a serving result under an audit-null regime, not under-powered shadows. On ogbn-products ($\sim 2.4M$ nodes) [28], Table XIX reports a *serving* probe on a 16 GB host plus BFS-subgraph LiRA at $n_{sh}=2$ (3 seeds) and $n_{sh}=4$ (5 seeds). Full-graph products LiRA OOMs on this host; we do *not* claim a million-node privacy audit. Both budgets keep LiRA near chance while HARP recovers Acc vs. strong LBP (≈ 0.69 vs. ≈ 0.35).

If audit LiRA ≈ 0.5 , prefer Frac=0; if LiRA $\gg 0.5$ or veracity requires ExactFrac $\geq c$, set Frac on val Acc (Frac=0.2 maximizes Acc on Cora/Chameleon; Frac=0.4 is the locked compliance default with 60% exact responses).

TABLE XX
CANARIES (LEFT) AND OGBN-ARXIV VOLUME (RIGHT; 3 SEEDS).
LiRA_{2/4} ARE DEFENSE-AWARE; LiRA₁₆ IS UNDEFENDED CREDIBILITY (NEAR-CHANCE).

(a) Cora canaries				(b) ogbn-arxiv			
	LiRA	Top	Plt.	ECE	Acc	LiRA ₂	LiRA ₄
Def.	0.568	0.551	0.527	0.052	0.666	0.502	0.503
none	0.517	0.548	0.543	0.146	0.666	—	0.510*
LBP	0.534	0.266	0.423	0.037	0.179	0.498	0.501
HARP				HARP	0.599	0.514	0.505

*Undefended LiRA at $n_{sh}=16$ (credibility); still near chance.

a) *Systems microbenchmarks.*: Cora setup ≈ 41 ms; Mass / release ≈ 0.51 ms vs. LBP ≈ 0.71 ms; products $\approx 3.3 \times 10^3$ QPS at ≈ 7.4 GB RSS.

b) *Operator workflow.*: Audit with defense-aware LiRA; log Mass/Frac/ECE/unprot-conf. If LiRA ≈ 0.5 , set Frac=0; else select Frac on validation Acc (Table VII), enforce session budget $B=1$ by default (Table X), and use GAP when a legal ϵ is required [13]. On Cora/Chameleon, val Acc selection recovers the Acc-maximizing Frac in the grid (Frac=0.2 matches the test oracle); LiRA or ExactFrac constraints instead pick a higher Frac on the same Pareto frontier. Re-audit after model or graph updates.

F. Protocol limitations

Equal-mass and MemGuard can beat HARP on Acc–LiRA when ExactFrac=0 is allowed; selective masking raises LiRA; multi-query needs $B=1$ for a one-shot guarantee; adaptive conf AUROC remains elevated on residual clean nodes even at Frac=0.60 (Table XVII); HARP is not DP; ogbn-products full LiRA needs more RAM/distributed shadows.

VIII. DISCUSSION

A. What HARP is for

HARP is for score-serving teams that must meet ExactFrac $\geq c > 0$ (replay-stable majority) while retaining usable Acc and an auditable noise spend. Membership audits decide whether Frac should be zero or positive and whether CFS can meet τ ; Acc and ECE measure utility/fidelity cost. When LiRA ≈ 0.5 and policy does not force strong σ , set Frac=0. When ExactFrac $\geq c$ is mandatory, selective release is the feasible class; uniform LBP is not (Remark 1).

B. Operational implications

Hop expansion is a coverage constraint matched to receptive field; Mass/Frac are release-side spend dials correlated with distortion (Table XV). ExactFrac enables content-addressable score caches that fresh uniform noise cannot (Tables XVI, XIV). Constructor choice is secondary to selectivity for ExactFrac; use audit-guided ranking when shadows are affordable and report Spearman quality (Table XIII).

C. Threats to validity

Primary Cora tables use five seeds with $\text{mean} \pm \text{SE}$; some secondary cells retain three seeds. Products LiRA is a BFS probe at $n_{\text{sh}} \leq 4$ —not a full 2.4M-node privacy audit; ogbn-arxiv undefended LiRA at $n_{\text{sh}} = 16$ remains near chance. MemGuard is hardened iterative search (not a line-by-line Jia reproduction); GAP-agg is aggregation-Laplace with analytical ε , not the full GAP codebase. Session budgets assume authenticated principals (no Sybil claim).

D. Relation to formal differential privacy

Remark 3: $\text{ExactFrac} > 0$ implies infinite global ε . Use GAP for regulatory DP; use HARP when $\text{ExactFrac} \geq c$ matters.

E. What we do not claim

Universal LiRA dominance; DP; Sybil-proof B ; that unprotected nodes are membership-safe; that topology ranking is always best.

F. Future work

Learned constructors, distributed full-graph LiRA, co-optimizing formal ε with Mass, and production serving traces.

IX. CONCLUSION

We proposed HARP, an ExactFrac -constrained score-serving framework for GNN prediction APIs. Fresh uniform perturbation is infeasible whenever a replay-stable majority is required; HARP keeps $\text{ExactFrac} \geq c$ by protecting a hop-expanded subset, exposes pluggable constructors and CFS under optional audit caps, and logs Mass/Frac/ B . Evidence emphasizes constrained feasibility, fidelity, and serving cost—with membership audits as deployment dials and residual clean-slice risk reported explicitly. Use training-time DP when a legal global ε is required.

a) *Reproducibility.*: Code (`defenses/harp.py`, `run_harp_constrained_upgrade.py`, `run_harp_eval.py`), locked configs, and frozen CSVs: <https://github.com/krishharma/privacy-gnn>. Runs used Darwin arm64 CPU; LiRA $n_{\text{shadows}} = 4$ primary.

REFERENCES

- [1] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, “Membership inference attacks against machine learning models,” in *Proc. IEEE Symp. Security Privacy (S&P)*, 2017, pp. 3–18.
- [2] A. Salem, Y. Zhang, M. Humbert, P. Berrang, M. Fritz, and M. Backes, “ML-Leaks: Model and data independent membership inference attacks and defenses on machine learning models,” in *Proc. NDSS*, 2019.
- [3] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, “Privacy risk in machine learning: Analyzing the connection to overfitting,” in *Proc. IEEE CSF*, 2018, pp. 268–282.
- [4] N. Carlini, S. Chien, M. Nasr, A. Song, T. Terzis, and F. Tramèr, “Membership inference attacks from first principles,” in *Proc. IEEE S&P*, 2022, pp. 1897–1914.
- [5] I. E. Olatunji, W. Nejdl, and M. Khosla, “Membership inference attack on graph neural networks,” arXiv:2101.06570, 2021.
- [6] J. Jia, A. Salem, M. Backes, Y. Zhang, and N. Z. Gong, “MemGuard: Defending against black-box membership inference attacks via adversarial examples,” in *Proc. ACM CCS*, 2019, pp. 259–274.
- [7] Y. Wu, D. Li, B. Chen, and L. Shen, “Adapting membership inference attacks to GNN for graph classification,” in *Proc. IEEE ICDM*, 2021, pp. 1429–1434.
- [8] Y. Zhang, Y. Zhao, Z. Li, X. Cheng, Y. Wang, O. Kotevska, P. S. Yu, and T. Derr, “A survey on privacy in graph neural networks: Attacks, preservation, and applications,” arXiv:2308.16375, 2023.
- [9] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *Proc. ICLR*, 2017.
- [10] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *Proc. NeurIPS*, 2017.
- [11] Y. Rong, W. Huang, T. Xu, and J. Huang, “DropEdge: Towards deep graph convolutional networks on node classification,” in *Proc. ICLR*, 2020.
- [12] C. A. Choquette-Choo, F. Tramèr, N. Carlini, and N. Papernot, “Label-only membership inference attacks,” in *Proc. ICML*, 2021, pp. 1964–1974.
- [13] S. Sajadmanesh et al., “GAP: Differentially private graph neural networks with aggregation perturbation,” in *Proc. USENIX Security*, 2023, pp. 3229–3246.
- [14] M. Abadi et al., “Deep learning with differential privacy,” in *Proc. ACM CCS*, 2016, pp. 308–318.
- [15] M. Nasr, R. Shokri, and A. Houmansadr, “Machine learning with membership privacy using adversarial regularization,” in *Proc. ACM CCS*, 2018, pp. 634–646.
- [16] M. Nasr, R. Shokri, and A. Houmansadr, “Comprehensive privacy analysis of deep learning,” in *Proc. IEEE S&P*, 2019, pp. 739–753.
- [17] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. ICML*, 2017, pp. 1321–1330.
- [18] J. Zhu et al., “Beyond homophily in graph neural networks: Current limitations and effective designs,” in *Proc. NeurIPS*, 2020.
- [19] H. Pei, B. Wei, K. C.-C. Chang, Y. Lei, and B. Yang, “Geom-GCN: Geometric graph convolutional networks,” in *Proc. ICLR*, 2020.
- [20] D. Lim, F. Hohne, X. Li, S. L. Huang, V. Gupta, O. Bhalerao, and S. N. Lim, “Large scale learning on non-homophilous graphs: New benchmarks and strong simple methods,” in *Proc. NeurIPS*, 2021.
- [21] P. Niu, C. Pan, S. Chen, and O. Milenković, “Graph Transductive Defense: a Two-Stage Defense for Graph Membership Inference Attacks,” arXiv:2406.07917, 2024.
- [22] C. Chen, X. Zhang, H. Qiu, J. Lou, Z. Liu, and X. Chen, “MaskArmor: Confidence masking-based defense mechanism for GNN against MIA,” *Information Sciences*, vol. 669, 2024.
- [23] M. A. P. Chamikara et al., “Can diffusion-based posterior smoothing secure GNNs from membership inference attacks?” in *Proc. AICCSA*, 2025.
- [24] V. Duddu, A. Boutet, and V. Shejwalkar, “Quantifying privacy leakage in graph embedding,” in *Proc. WSDM*, 2020, pp. 76–84.
- [25] M. Fredrikson, S. Jha, and T. Ristenpart, “Model inversion attacks that exploit confidence information,” in *Proc. ACM CCS*, 2015, pp. 1322–1333.
- [26] F. Tramèr et al., “Stealing machine learning models via prediction APIs,” in *Proc. USENIX Security*, 2016, pp. 601–618.
- [27] C. Dwork, “Differential privacy,” in *Proc. ICALP*, 2006, pp. 1–12.
- [28] W. Hu et al., “Open graph benchmark: Datasets for machine learning on graphs,” in *Proc. NeurIPS*, 2020.
- [29] P. Veličković et al., “Graph attention networks,” in *Proc. ICLR*, 2018.
- [30] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, “How powerful are graph neural networks?” in *Proc. ICLR*, 2019.
- [31] I. Mironov, “Rényi differential privacy,” in *Proc. IEEE CSF*, 2017, pp. 263–275.
- [32] B. Jayaraman and D. Evans, “Evaluating differentially private machine learning in practice,” in *Proc. USENIX Security*, 2019, pp. 1895–1912.
- [33] C. Song, T. Ristenpart, and V. Shmatikov, “Machine learning models that remember too much,” in *Proc. ACM CCS*, 2017, pp. 587–601.
- [34] A. Niculescu-Mizil and R. Caruana, “Predicting good probabilities with supervised learning,” in *Proc. ICML*, 2005, pp. 625–632.
- [35] Y. Jiang et al., “Unveiling privacy vulnerabilities: Investigating the role of structure in graph data,” in *Proc. ACM KDD*, 2024. (GPS: structure→attribute leakage; GHRatio; learnable edge sampling.)
- [36] Q. Yuan et al., “Node-Aware Differentially Private Graph Neural Networks for Graph Representation Learning,” arXiv:2308.04943, 2023.
- [37] M. Aerni, J. Zhang, and F. Tramèr, “Evaluations of machine learning privacy defenses are misleading,” in *Proc. ACM CCS*, 2024.
- [38] X. He, R. Wen, Y. Wu, M. Backes, Y. Shen, and Y. Zhang, “Node-level membership inference attacks against graph neural networks,” arXiv:2102.05429, 2021.
- [39] M. Khosla, “Impact of graph structure on membership-inference risk for graph neural networks,” *Proc. Priv. Enhancing Technol. (PoPETs)*, 2026.